



Design Report

Electric Petrol

Infrastructure Upgrade Design Report

Midleton Branch Site - Clustered Shared Storage, High Availability & AD Domain
Tree Integration

Assignment 3 - HCT

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Site: Midleton (Site 50)

IP Allocation: 10.50.0.0/16

Domain: Midleton.ads.electric-petrol.ie

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1.0	01/04/2026	L00187595	Initial draft
1.1	02/04/2026	L00187595	Server specification and storage design
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Executive Summary

This design report presents the technical plan for upgrading the Electric Petrol Midleton branch site (Site 50, IP range 10.50.0.0/16, domain Midleton.ads.electric-petrol.ie). The project addresses three key requirements:

- Add a third virtualization server at the Midleton site and implement Clustered Shared Storage
- Ensure the site continues to operate in the event of a single server failure
- Reengineer AAA to be part of the domain tree for electric-petrol.ie (as Midleton.ads.electric-petrol.ie)

The existing infrastructure at Midleton consists of two Dell R440 servers, each with 256GB RAM in 16GB DIMMs, 2 x 240GB SSD in RAID 1 on a BOSS card, 4 x 1TB HDDs in RAID5, 2 x 1GB/s onboard Ethernet, and 2 x 10GB/s Ethernet in PCI-e slots. Each server runs Windows Server 2019 Datacentre edition with VMs for domain controller (DCx, 4GB), file server (FSx, 8GB), database server with MSSQL (DBx, 24GB), and Ubuntu 2004 Syslog (SLx, 8GB). A Galleon NTS-6002-GPS time server is available via Ethernet.



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1. Equipment Audit of Existing Midleton Site

In scenario 1, at the Midleton branch site (Site 50), we installed host servers and used virtualization to build a simple, local solution, appropriate to the size of the business. We installed two host servers and used virtualization to build a simple appropriate solution.

1.1 Existing Server Hardware Specification

The servers were Dell R440s with:

- 256GB in 16GB DIMMs
- 2 x 240GB SSD in RAID 1 on a BOSS card for the operating system
- 4 x 1TB HDDs in RAID5 for data
- 2 x 1GB/s on board Ethernet cards
- 2 x 10GB/s Ethernet cards in a PCI-e slot

• Component	• Specification
• Memory	• 256GB in 16GB DIMMs
• OS Storage	• 2 x 240GB SSD in RAID 1 on a BOSS card
• Data Storage	• 4 x 1TB HDDs in RAID5
• Onboard Networking	• 2 x 1GB/s on board Ethernet cards
• Additional Networking	• 2 x 10GB/s Ethernet cards in a PCI-e slot
• Time Server	• Galleon NTS-6002-GPS available via Ethernet

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1.2 Existing Virtual Machine Configuration

Each host server was configured with Windows 2019 Server (Datacentre edition) and has the following VMs:

VM Name	Operating System	Role	RAM
DCx	Windows 2019 Server	Domain Controller	4GB
FSx	Windows 2019 Server	File Server	8GB
DBx	Windows 2019 Server	Database Server (MSSQL)	24GB
SLx	Ubuntu 2004	Syslog Server	8GB

Total RAM utilised per host: 44GB out of 256GB available (17.2% utilisation).

There is a Galleon NTS-6002-GPS time server available via Ethernet.



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1.3 Midleton Site Network Allocation

The Midleton branch site (Site 50) has been allocated the IP range 10.50.0.0/16 with the following VLAN structure:

VLAN	Purpose	Network
1	ISP	10.50.1.0/24
2	Hosts	10.50.2.0/24
3	Staff Radio	10.50.3.0/24
4	Power Infrastructure	10.50.4.0/24
5	Technical Clients	10.50.5.0/24
6	Infrastructure Servers	10.50.6.0/24
7	Telephony	10.50.7.0/24
8	HVAC Infrastructure	10.50.8.0/24
9	Storage	10.50.9.0/24
10	NetMan	10.50.10.0/24
11	Security Cameras	10.50.11.0/24
12	Server OOBM	10.50.12.0/24

1.4 Midleton Site IP Address Assignments

The following fixed IP addresses are assigned for key infrastructure on the Midleton site:

Device	VLAN	IP Address
Host Server 1 (Host11)	VLAN 2 - Hosts	10.50.2.21
Host Server 2 (Host12)	VLAN 2 - Hosts	10.50.2.22
Domain Controller 1 (DC1)	VLAN 6 - Infra Servers	10.50.6.31
Domain Controller 2 (DC2)	VLAN 6 - Infra Servers	10.50.6.32
Storage Server 1	VLAN 6 - Infra Servers	10.50.6.41
Storage Server 2	VLAN 6 - Infra Servers	10.50.6.42
Database Server 1 (DB1)	VLAN 6 - Infra Servers	10.50.6.51
Database Server 2 (DB2)	VLAN 6 - Infra Servers	10.50.6.52
Storage 1 (iSCSI)	VLAN 9 - Storage	10.50.9.41
Storage 2 (iSCSI)	VLAN 9 - Storage	10.50.9.42



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2. Technical Tasks Required for the Upgrade

This section details the exact technical tasks for upgrading the Midleton branch site (Site 50, 10.50.0.0/16).

2.1 Work Packages Overview

WP	Task	Description
WP1	Server Procurement	Procure and install Dell PowerEdge R450 (third server) at Midleton site
WP2	Storage Deployment	Deploy Dell PowerVault ME5024 shared storage array at Midleton
WP3	Existing Server Upgrades	Add HBA controllers and reconfigure 10GbE NICs for iSCSI on both Dell R440s
WP4	Cluster Configuration	Configure VMware HA, DRS, and Clustered Shared Volumes
WP5	VM Migration	Migrate all VMs from local to shared storage using Storage vMotion
WP6	AD Restructuring	Migrate standalone AD to Midleton.ads.electric-petrol.ie child domain
WP7	Testing	Validate failover, authentication, and performance

2.2 Implications and Considerations

Downtime: Each existing Dell R440 server (Host11 at 10.50.2.21, Host12 at 10.50.2.22) requires a 2-4 hour maintenance window for HBA controller installation. VM migration via Storage vMotion is non-disruptive.

Licensing: Windows Server 2019 Datacentre edition licenses cover unlimited VMs per host. VMware vSphere Enterprise Plus licensing is required for HA and DRS.

Network: The existing 2 x 10GB/s Ethernet cards in each PCI-e slot will be dedicated to iSCSI storage traffic on VLAN 9 (Storage, 10.50.9.0/24). Storage IPs: 10.50.9.41 and 10.50.9.42.

Backup: Full backups of all VMs (DCx 4GB, FSx 8GB, DBx 24GB MSSQL, SLx 8GB Ubuntu 2004) must be taken before any hardware changes.



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3. Upgrades for Existing Servers

Both existing Dell R440 servers at Midleton (Host11 at 10.50.2.21 and Host12 at 10.50.2.22), each with 256GB RAM, 2 x 240GB SSD BOSS, 4 x 1TB HDD RAID5, 2 x 1GB/s + 2 x 10GB/s Ethernet, require the following upgrades:

3.1 Hardware Upgrades

Component	Current (Dell R440)	Change Required	Justification
Storage HBA	None (onboard PERC)	Add Dell HBA330 SAS/SATA HBA	Shared SAN/iSCSI storage connectivity
10GbE NICs	2 x 10GB/s PCI-e (general)	Dedicate for iSCSI with MPIO on VLAN 9 (10.50.9.0/24)	Isolated high-speed storage network
Memory	256GB in 16GB DIMMs	No change	Only 44GB of 256GB used per host
Local Storage	4 x 1TB HDDs in RAID5	Retain for scratch/ISOs	VM storage moves to shared SAN
OS Storage	2 x 240GB SSD RAID 1 (BOSS)	No change	BOSS card boot config unchanged

3.2 Software Upgrades

- Update VMware ESXi to latest 7.0 U3 patch for HA compatibility
- Configure iSCSI Software Adapter on VLAN 9 (Storage, 10.50.9.0/24)
- Add Storage VLAN 9 with MTU 9000 on virtual switches
- Reconfigure NTP to sync with Galleon NTS-6002-GPS time server
- Rejoin ESXi hosts to Midleton.ads.electric-petrol.ie domain



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4. Specification of the New Third Server

The third server for the Midleton site must be compatible with the existing Dell R440 infrastructure. The Dell PowerEdge R450 is the recommended successor.

4.1 Dell PowerEdge R450 Specification

Component	Specification	Notes
Model	Dell PowerEdge R450 (1U Rack)	Direct successor to Dell R440
Processor	2 x Intel Xeon Silver 4314 (16C/32T)	Matches/exceeds existing R440
Memory	256GB DDR4 in 16GB DIMMs	Matches existing (256GB in 16GB DIMMs)
OS Storage	2 x 480GB SSD RAID 1 on BOSS-S2	Upgraded from 2 x 240GB SSD BOSS
Local Data Storage	4 x 1TB SAS HDDs in RAID5	Matches existing (4 x 1TB HDDs RAID5)
Storage HBA	Dell HBA355i SAS/SATA	Pre-installed for shared SAN/iSCSI
Onboard Networking	2 x 1GB/s Ethernet	Matches existing (2 x 1GB/s onboard)
Additional Networking	2 x 10GB/s Ethernet PCI-e	Matches existing; dedicated to iSCSI on VLAN 9
Power Supply	2 x 600W redundant hot-swap PSU	Hardware fault tolerance

4.2 New Server Network Configuration

The third server (Host13) will be assigned the following IPs on the Midleton network (10.50.0.0/16):

Interface	VLAN	IP Address
Management NIC 1	VLAN 2 - Hosts	10.50.2.23
Management NIC 2	VLAN 10 - NetMan	10.50.10.23
iSCSI NIC 1	VLAN 9 - Storage	10.50.9.43
iSCSI NIC 2	VLAN 9 - Storage	10.50.9.44



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4.3 VM Distribution Across Three Hosts

With three servers, VMs are distributed for redundancy with anti-affinity rules:

Host	VMs	RAM Used	Key IPs (VLAN 6)
Host11 (Dell R440) - 10.50.2.21	DC1 (4GB), FS1 (8GB)	12GB/256GB	DC1: 10.50.6.31
Host12 (Dell R440) - 10.50.2.22	DB1 (24GB), SL1 (8GB)	32GB/256GB	DB1: 10.50.6.51
Host13 (Dell R450) - 10.50.2.23	DC2 (4GB), FS2 (8GB)	12GB/256GB	DC2: 10.50.6.32, DB2: 10.50.6.52
Failover	Any host absorbs full workload	88GB max/256GB	N+1 redundancy assured



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5. Clustered Shared Storage Implementation

Clustered Shared Volumes (CSV) provide shared storage accessible by all three ESXi hosts at the Midleton site simultaneously, enabling VM high availability and live migration.

5.1 Shared Storage Array

Component	Specification
Storage Array	Dell PowerVault ME5024 (2U, 24-bay SFF)
Drives	12 x 2.4TB 10K SAS HDDs in RAID 10 (~12TB usable)
Cache	2 x 4GB read/write cache per controller
Connectivity	4 x 10GbE iSCSI (2 per controller, dual controllers)
RAID	RAID 10 for performance and redundancy
Protocol	iSCSI over 10GbE

5.2 Storage Network Design - Midleton VLAN 9

The dedicated Storage VLAN 9 (10.50.9.0/24) isolates iSCSI traffic at the Midleton site. Each host's 2 x 10GB/s Ethernet PCI-e cards are configured for iSCSI multipathing:

Device	VLAN 9 IP (10.50.9.x)	Role
Host11 (Dell R440) - Path A	10.50.9.21	iSCSI Initiator Path 1
Host11 (Dell R440) - Path B	10.50.9.22	iSCSI Initiator Path 2
Host12 (Dell R440) - Path A	10.50.9.23	iSCSI Initiator Path 1
Host12 (Dell R440) - Path B	10.50.9.24	iSCSI Initiator Path 2
Host13 (Dell R450) - Path A	10.50.9.43	iSCSI Initiator Path 1
Host13 (Dell R450) - Path B	10.50.9.44	iSCSI Initiator Path 2
Storage Array Controller A/B	10.50.9.41 / 10.50.9.42	iSCSI Target (dual controller)

All iSCSI ports use MTU 9000 (jumbo frames) with Round Robin multipathing policy.

5.3 VMFS Datastore Layout

Datastore Name	Size	Purpose
Midleton-DS-Production	6TB	Production VMs (DC, FS, DB, Syslog)
Midleton-DS-Templates	2TB	VM templates, ISOs, deployment images
Midleton-DS-Backup	4TB	VM snapshots and backup storage



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5.4 VM Migration to Shared Storage

Existing VMs are migrated from local datastores (4 x 1TB HDDs RAID5 per R440) to shared storage using Storage vMotion (non-disruptive):

- Phase 1: Migrate DC1, DC2 (Windows 2019, 4GB each) - smallest, lowest risk
- Phase 2: Migrate SL1 (Ubuntu 2004 Syslog, 8GB)
- Phase 3: Migrate FS1, FS2 (Windows 2019 File Server, 8GB each)
- Phase 4: Migrate DB1 (Windows 2019 MSSQL, 24GB) during maintenance window
- Phase 5: Verify all VMs on shared storage; reclaim local datastores



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6. High Availability Design

The Midleton site must continue operating during a single server failure. VMware vSphere HA combined with Clustered Shared Storage on VLAN 9 (10.50.9.0/24) achieves this.

6.1 vSphere HA Configuration

- Admission Control: Reserve resources for 1 host failure (33% reservation)
- Host Monitoring: Heartbeat datastores on Midleton shared storage
- VM Restart Priority: High for DC1/DC2 and DB1, Medium for FS1/FS2, Low for SL1
- VM Monitoring: Heartbeat via VMware Tools
- Host Isolation Response: Power off and restart VMs on surviving hosts
- Network Redundancy: Management on dual 1GB/s NICs with NIC teaming on VLAN 2 (10.50.2.0/24)

6.2 DRS Configuration

- Automation Level: Fully Automated
- Migration Threshold: Level 3 (moderate)
- Anti-Affinity: DC1 (10.50.6.31) and DC2 (10.50.6.32) never on same host
- Anti-Affinity: DB1 (10.50.6.51) and DB2 (10.50.6.52) never on same host
- Live migration via vMotion over VLAN 2 (10.50.2.0/24) for zero-downtime rebalancing

6.3 Failover Capacity Analysis

Scenario	Available RAM	Status
Normal (3 hosts at Midleton)	768GB total, 88GB used (11.5%)	Fully operational with DRS
Single Host Failure (2 hosts)	512GB total, 88GB used (17.2%)	All VMs operational via HA
Two Host Failure (1 host)	256GB total, 88GB used (34.4%)	All VMs on single host (degraded)

Each host has 256GB RAM (in 16GB DIMMs). With only 44GB utilised per VM set, any single host can absorb the full workload (VMware, 2024).



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7. New Active Directory Domain Tree Design

The Midleton branch currently operates a standalone AD domain. It must be reengineered to join the electric-petrol.ie domain tree as Midleton.ads.electric-petrol.ie.

7.1 Current AAA Configuration

- Windows 2019 domain controller (DCx) using 4GB RAM - standalone domain
- ESXi hosts (10.50.2.21, 10.50.2.22) joined to local AD
- vCenter Server joined to local AD for VM management
- Ubuntu 2004 Syslog server (SLx, 8GB) collecting security audit logs

7.2 Target Domain: Midleton.ads.electric-petrol.ie

Domain Tree Structure:

electric-petrol.ie (Forest Root - HQ, 10.0.0.0/16)

|-- ads.electric-petrol.ie (AD Services)

|-- Midleton.ads.electric-petrol.ie (Midleton Branch, 10.50.0.0/16)

- Forest Root: electric-petrol.ie at HQ (10.0.0.0/16)
- Child Domain: Midleton.ads.electric-petrol.ie (10.50.0.0/16)
- Trusts: Automatic two-way transitive trusts
- Global Catalog: DC1 (10.50.6.31) and DC2 (10.50.6.32) both as GC servers
- DNS: AD-integrated DNS for Midleton.ads.electric-petrol.ie zone
- Time: PDC Emulator syncs to Galleon NTS-6002-GPS; other DCs sync from PDC

7.3 Domain Controller Deployment

DC	Host / IP	Roles	RAM
DC1	Host11 (10.50.2.21) / DC IP: 10.50.6.31	AD DS, DNS, GC, FSMO roles	4GB
DC2	Host13 (10.50.2.23) / DC IP: 10.50.6.32	AD DS, DNS, GC (replica)	4GB
Anti-Affinity	DRS Enforced	DC1 and DC2 never on same host	-

7.4 Migration Steps

1. Deploy forest root electric-petrol.ie at HQ (10.0.0.0/16)
2. Establish VPN/WAN between Midleton (10.50.0.0/16) and HQ
3. Promote DC1 (10.50.6.31) as first child DC for Midleton.ads.electric-petrol.ie
4. Promote DC2 (10.50.6.32) as replica child DC on Host13



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5. Migrate users/groups/computers via ADMT from standalone to Midleton.ads.electric-petrol.ie
6. Update Group Policy Objects for new domain
7. Rejoin ESXi Host11 (10.50.2.21), Host12 (10.50.2.22), Host13 (10.50.2.23) to new domain
8. Rejoin vCSA to Midleton.ads.electric-petrol.ie
9. Configure Syslog SLx (Ubuntu 2004, 8GB) to receive audit logs from new domain
10. Decommission old standalone DC after validation

7.5 Security Policies

- Password Policy: 12+ characters, complexity, 90-day rotation
- Account Lockout: 5 failed attempts, 30-minute lockout
- Kerberos: Domain tree trusts enable single sign-on across electric-petrol.ie
- Audit: Events forwarded to Ubuntu 2004 Syslog server (SLx, 8GB) at Midleton
- RADIUS/NPS: Centralised RADIUS on DCs for network equipment auth



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8. Implementation Management Plan

This section outlines the phased implementation plan for the Midleton site (10.50.0.0/16) infrastructure upgrade.

8.1 Phase 1: Hardware Deployment

Task	Details	Duration
Rack and cable Host13 (Dell R450)	Install in Midleton server room, connect 2x1GB/s + 2x10GB/s	1 day
Install Dell PowerVault ME5024	Rack storage array, cable iSCSI to VLAN 9 (10.50.9.0/24)	1 day
Configure iSCSI network	Storage1: 10.50.9.41, Storage2: 10.50.9.42, host paths: .21-.24,.43-.44	1 day
Verify Galleon NTS-6002-GPS	Confirm NTP via Ethernet to all hosts	0.5 day

8.2 Phase 2: VMware Configuration

Task	Details	Duration
Install ESXi 7.0 on Host13	Configure mgmt IP 10.50.2.23 on VLAN 2	0.5 day
Add Host13 to vCenter cluster	Join Midleton-Cluster, configure DRS/HA	0.5 day
Create shared datastores	Midleton-DS-Production (6TB), Templates (2TB), Backup (4TB)	1 day
Storage vMotion all VMs	Migrate DC1, DC2, SL1, FS1, FS2, DB1 to shared storage	2 days
Configure HA/DRS rules	Anti-affinity: DC1/DC2, DB1/DB2; admission control 33%	0.5 day

8.3 Phase 3: Active Directory Reengineering

Task	Details	Duration
Establish WAN to HQ	VPN between Midleton 10.50.0.0/16 and HQ 10.0.0.0/16	1 day
Promote DC1 to child domain	Midleton.ads.electric-petrol.ie, DC1 at 10.50.6.31	1 day
Promote DC2 as replica	DC2 at 10.50.6.32 on Host13, GC enabled	0.5 day
Migrate users/groups via ADMT	All accounts to Midleton.ads.electric-petrol.ie	2 days



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Rejoin ESXi hosts to new domain	Host11 (10.50.2.21), Host12 (10.50.2.22), Host13 (10.50.2.23)	1 day
Update vCSA and Syslog config	Rejoin vCSA; reconfigure SLx for new domain events	0.5 day

8.4 Phase 4: Testing and Validation

Task	Details	Duration
HA failover test	Simulate host failure, verify VM restart on surviving hosts	0.5 day
DRS rebalancing test	Verify anti-affinity and live migration	0.5 day
AD replication test	Verify DC1-DC2 replication, DNS resolution for Middleton.ads.electric-petrol.ie	0.5 day
Storage failover test	Simulate iSCSI path failure, verify multipath failover	0.5 day
Full site failover	Power off one host, confirm all services operational	1 day



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9. Risk Assessment

The following risk assessment identifies key risks to the Midleton infrastructure upgrade project.

Risk	Likelihood	Impact	Mitigation	Residual Risk
Single host failure during migration	Medium	High	Schedule Storage vMotion in maintenance window; keep local copies until verified	Low
iSCSI storage network failure	Low	Critical	Dual controller (10.50.9.41/42), multipath with 2 paths per host, jumbo frames	Low
AD migration data loss	Low	High	Full AD backup before migration; ADMT rollback capability	Low
DNS resolution failure post-migration	Medium	High	Keep old DNS records during transition; conditional forwarders to HQ	Low
WAN link failure to HQ	Medium	Medium	Midleton DCs cache credentials locally; site operates autonomously	Low
Hardware delivery delay for Host13	Medium	Medium	Order Dell R450 early; use temporary hardware if needed	Medium



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10. Conclusion

This report presents a comprehensive infrastructure upgrade plan for the Midleton (Site 50) branch of Electric Petrol, addressing all three customer requirements:

- Requirement 1: A third virtualisation host (Dell R450, Host13 at 10.50.2.23) with Dell PowerVault ME5024 shared storage on VLAN 9 (10.50.9.0/24) provides Clustered Shared Volumes accessible by all three hosts ([Dell Technologies, 2024](#)).
- Requirement 2: VMware vSphere HA with DRS ensures the Midleton site continues operating during any single server failure. With 768GB total RAM and only 88GB utilised, any single host can absorb the full workload ([VMware, 2024](#)).
- Requirement 3: Active Directory has been reengineered from a standalone domain to Midleton.ads.electric-petrol.ie, integrating into the electric-petrol.ie forest with two-way transitive trusts, dual domain controllers (DC1 at 10.50.6.31, DC2 at 10.50.6.32), and centralised authentication ([Microsoft, 2024](#)).

The phased implementation plan minimises disruption to Midleton operations while delivering enterprise-grade redundancy, scalability, and security for the Electric Petrol network (Dell Technologies, 2024; VMware, 2024; Microsoft, 2024).



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Appendix A: Equipment Audit Summary

Source: Equipment Audit of Existing Sites (Electric Petrol)

Physical Server Specification (Dell R440):

- RAM: 256 GB in 16 GB DIMMs
- Boot: 2 x 240 GB SSD in a RAID 1 BOSS configuration
- Data: 4 x 1 TB HDD in a RAID 5 configuration
- Network: 2 x 1 GB/s onboard Ethernet
- Network: 2 x 10 GB/s PCI-e Ethernet
- Hypervisor: VMware ESXi (version audited)

Virtual Machine Inventory per Host:

VM	OS	RAM	Role
DCx	Windows 2019	4GB	Domain Controller
FSx	Windows 2019	8GB	File Server
DBx	Windows 2019 + MSSQL	24GB	Database Server
SLx	Ubuntu 2004	8GB	Syslog Server

Time Server:

- Galleon NTS-6002-GPS with Ethernet connection
- Provides NTP to all hosts and VMs at the Middleton site

Appendix B: References

- Dell Technologies (2024) PowerEdge R440 Technical Guide. Available at: <https://www.dell.com/support/home/en-ie/product-support/product/poweredge-r440/docs>
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